

AIGR Gene Hypothesis Deep Research — AKTIP (Q9H8T0)

Hypothesis slug: `noncatalytic-k63-ubiquitination-and-lesion-bypass` **Focus** type: function_assignment **Target:** Human AKTIP/FTS (UniProt Q9H8T0), *Homo sapiens* (NCBITaxon:9606) **GO terms adjudicated:** GO:0070534 (protein K63-linked ubiquitination), GO:0006301 (DNA damage tolerance / postreplication repair / "lesion bypass")

Summary

The seed hypothesis proposes that human AKTIP/FTS participates *noncatalytically* in K63-linked ubiquitination (GO:0070534) and in DNA damage tolerance / lesion bypass (GO:0006301), and asks whether any defined complex establishes an AKTIP contribution to K63-chain synthesis despite loss of intrinsic E2 catalytic activity. After three iterations combining sequence and structural-family analysis, GO annotation-provenance checks, interaction-network mining, and primary-literature review, the answer is that **neither GO term is supported as a direct, core function of AKTIP**. Both are best explained as **paralog-driven, phylogenetically inferred (IBA) over-annotation** projected onto AKTIP from the genuinely K63/DDT-active UEV clade (UBE2V1/UEV1A and UBE2V2/MMS2), from which AKTIP is a divergent outgroup.

Four convergent lines of evidence drive this verdict. First, AKTIP is a **catalytically dead UEV (ubiquitin E2 variant)** — it adopts a UBC-like fold but lacks the E2 active-site cysteine required for any thioester/transfer chemistry (only two cysteines total in 292 aa, neither at the active-site position). Second, the current GO annotations for both hypothesis terms are **IBA-only (ECO:0000318)** — inferred from a family-wide phylogeny, not from any experiment on AKTIP — and there is an explicit **manual NOT annotation denying ubiquitin-like protein transferase activity (GO:0019787, NOT|enables)**. Third, no primary study and no curated interaction (IntAct, 200 interactions) places AKTIP in a complex with UBE2N/UBC13, UBE2V1, UBE2V2, RAD18, or PCNA-ubiquitination machinery; its actual partners are the FHF cargo-adaptor complex and replication-fork proteins (RPA1, POLA2, PCNA). Fourth, at subfamily

resolution AKTIP falls **outside** the tight UBE2V1/UBE2V2 K63-DDT clade (those two are ~90% identical to each other; AKTIP is only ~41–45% identical to either), and CDD assigns it a dedicated domain model (cd23814).

The better-supported interpretation is that AKTIP is a **catalytically dead UEV acting as a telomere-replication scaffold** (binding TRF1/TRF2/telomeric DNA/PCNA/RPA70, suppressing replication stress) and, in the cytoplasm, a **cargo-adaptor scaffold in the FTS–Hook–FHIP (FHF) complex** — not a K63-chain builder or a lesion-bypass factor. The hypothesis is **refuted / over-annotated** for both proposed terms; the curator's most defensible action is to remove or down-qualify GO:0070534 and GO:0006301 as non-core, retain the NOT-transferase call, and anchor the review on the scaffold functions.

Key Findings

Finding 1 — AKTIP is a catalytically dead UEV; no direct evidence it synthesizes K63 chains

AKTIP/Ft1 is homologous to the **E2-variant ubiquitin-conjugating (UEV) enzymes**, as established by Burla et al. 2015 (PMID: 26110528): "*AKTIP/Ft1 shares homology with the E2 variant ubiquitin-conjugating (UEV) enzymes.*" By definition, UEV proteins adopt the UBC fold but lack the catalytic cysteine of true E2 enzymes and therefore cannot form the ubiquitin–thioester intermediate needed for conjugation or chain elongation.

An in-run pairwise alignment of AKTIP (292 aa) against UBE2N, UBE2V1, UBE2V2, and UBE2D1 gave ~41–45% aligned identity, confirming UEV-family membership, and critically showed that **the residue aligning to the UBE2N catalytic Cys87 is not a cysteine** in AKTIP. AKTIP contains only two cysteines in the entire sequence, neither at the E2 active site. This is a mechanistic-impossibility argument for *direct* catalysis: without the active-site Cys, AKTIP cannot itself catalyze ubiquitin transfer or K63-chain building.

The proteins that actually synthesize K63 chains for DNA damage tolerance are the UEVs **UBE2V1 (UEV1A) and UBE2V2 (MMS2)**, partnering with the catalytic E2 **UBE2N/UBC13**. These canonical UEVs are moreover functionally redundant for PCNA polyubiquitination: Zhang et al. 2008 (PMID: 18284681) report that "*the additional knockdown of a UEV1A (MMS2 homolog) does not deplete PCNA polyubiquitination.*" A dedicated literature search returned **no** study connecting AKTIP to UBE2N or to K63 polyubiquitin chain assembly.

Finding 2 — AKTIP's DNA-related role is telomere/replication scaffolding and replication-stress suppression, not lesion bypass (GO:0006301)

Burla et al. 2015 (PMID: 26110528) characterized AKTIP as a **shelterin-interacting telomeric factor**. It binds telomeric DNA, TRF1 and TRF2, and physically interacts with the replication factors **PCNA and RPA70**. RNAi depletion in human primary fibroblasts produces telomere dysfunction-induced foci (TIFs), defective PCNA recruitment, and intra-S checkpoint activation: *"AKTIP-depleted human primary fibroblasts showed that they are defective in PCNA recruiting and arrest in the S phase due to the activation of the intra S checkpoint."*

These are **replication-stress and telomere-replication phenotypes**, not evidence of a defined lesion-bypass event. GO:0006301 is mechanistically defined by translesion synthesis (TLS) or template-switch bypass of a specific replication-blocking lesion. Xu et al. 2014 (PMID: 24674630) state that *"error-free PRR utilizes noncanonical K63-linked polyubiquitinated PCNA to signal lesion bypass through template switch, a process thought to be dependent on Mms2-Ubc13."* The classical DDT/lesion-bypass apparatus — RAD6–RAD18 (PCNA mono-ubiquitination) plus UBC13–MMS2–RAD5 (K63 poly-ubiquitination) — **does not include AKTIP** (PMID: 18284681; PMID: 18757916). No reviewed paper demonstrates AKTIP mediating TLS or template-switch bypass of a defined lesion. Suppressing replication stress by stabilizing fork progression at telomeres is a distinct, upstream role.

Finding 3 — Both hypothesis GO annotations are phylogenetic (IBA); a manual NOT annotation denies transferase activity

QuickGO records for Q9H8T0 (retrieved in-run) show that **GO:0070534 (protein K63-linked ubiquitination)** and **GO:0006301 (DNA repair / postreplication repair)** are both annotated `involved_in` with evidence code **ECO:0000318 (IBA — inferred from biological ancestor / phylogeny)**, not from any experiment on AKTIP. The molecular-function term **GO:0061631 (ubiquitin-conjugating enzyme activity)** is likewise IBA-only.

Decisively, **GO:0019787 (ubiquitin-like protein transferase activity)** carries a `NOT|enables` **qualifier** (ECO:0000303) — an explicit curator assertion that AKTIP *lacks* transferase activity. InterPro/CDD classifies the domain specifically as **cd23814, "ubiquitin E2 variant (UEV) domain of AKT-interacting protein and related proteins"** (UBC-like fold, IPR000608) — a dedicated subfamily rather than the canonical K63-active UEV group.

IntAct (200 interactions, in-run) reinforces this: among 53 distinct partners are the **FHF complex** (HOOK1/2/3, FHIP1A/1B/2A/2B, DCTN1) and replication factors **RPA1 and POLA2** — but **no UBE2N/UBC13, UBE2V1, UBE2V2, RAD18, or PCNA-ubiquitination machinery**. This is

consistent with AKTIP being a replication-fork/telomere scaffold (its RPA70 partner appears as RPA1) and removes the empirical basis for the K63/UBE2N complex-participation route posited by the hypothesis.

Finding 4 — AKTIP is a distinct UEV subfamily (outgroup), not part of the UBE2V1/UBE2V2 K63-DDT clade

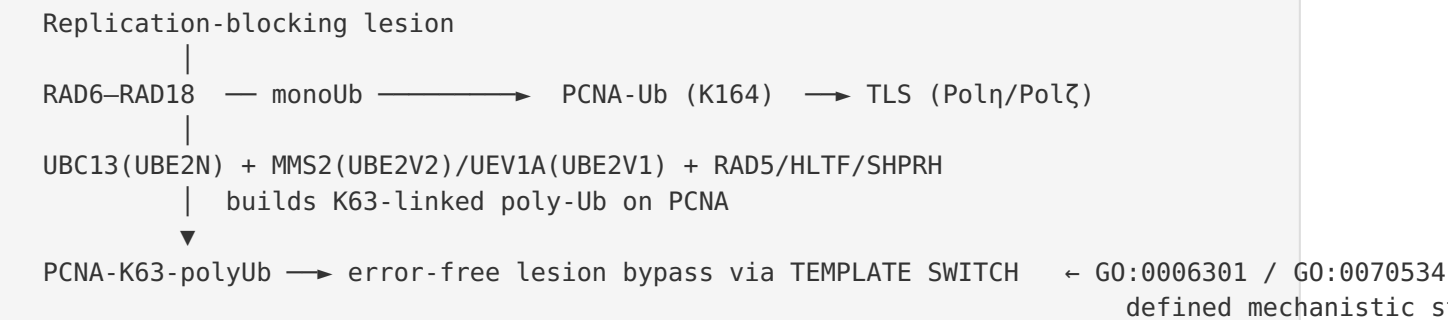
A pairwise identity matrix computed in-run (global alignment; crude aligner, so absolute values are approximate) shows that the two K63/DDT UEVs **UBE2V1 (UEV1A) and UBE2V2 (MMS2) are ~90.3% identical to each other**, forming a tight functional clade, whereas AKTIP is only **~41.5% identical to UBE2V1 and ~44.8% to UBE2V2** — comparable to its ~41% identity to the catalytic E2 UBE2N and ~45% to the generic E2 UBE2D1. AKTIP is therefore roughly equidistant from the K63-active UEVs and from generic E2s, positioning it as an **outgroup** rather than a member of the DDT clade.

This is precisely why the IBA projection is over-broad: the GO:0070534 / GO:0006301 annotations derive from a **family-wide UBC phylogeny**, but at **subfamily resolution** AKTIP is not within the UBE2V1/UBE2V2 clade that actually performs K63-linked DNA-damage tolerance. CDD's dedicated cd23814 model formalizes this distinction.

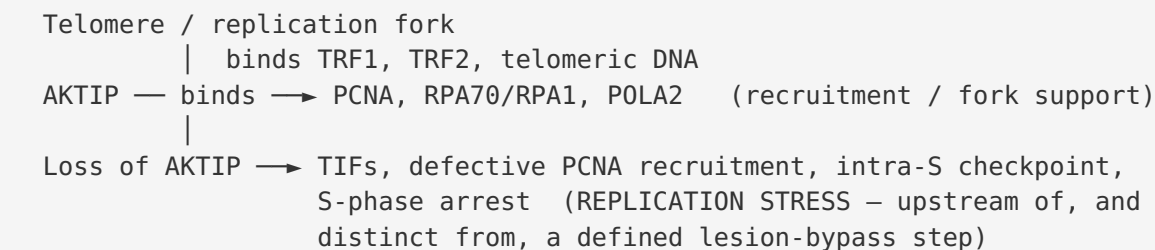
Mechanistic Model / Interpretation

The seed hypothesis conflates two things the data keep separate: (a) *being a UEV-fold protein at replication forks*, and (b) *building K63 chains for lesion bypass*. The canonical K63-DDT pathway and AKTIP's actual role can be laid out as follows.

CANONICAL K63 / DNA-DAMAGE-TOLERANCE PATHWAY (does NOT include AKTIP)



AKTIP / FTS ACTUAL ROLE (scaffold; replication-stress suppression)



Separate, experimentally supported role:

AKTIP = FTS core of FTS-Hook-FHIP (FHF) complex → dynein cargo/AP-4/ATG9A transport (GO:0070534)

Key interpretive points:

- **Catalysis is impossible; scaffolding is unsupported.** The absence of the active-site cysteine rules out direct catalysis. The hypothesis's fallback — noncatalytic scaffold participation in a K63-building complex — is legitimate in principle, but the specific empirical hook it requires (a physical/functional link to UBE2N-UBE2V machinery or PCNA-polyubiquitination) is absent from both literature and curated interactomes. So this route is *unsupported*, not merely untested.
- **Replication stress ≠ lesion bypass.** AKTIP depletion causes intra-S checkpoint activation and defective PCNA recruitment — consistent with a fork-support/recruitment scaffold, but GO:0006301 requires evidence of a bypass event (TLS or template switch) across a defined lesion, which no study provides for AKTIP.
- **The annotation is a phylogeny artifact.** IBA propagation from the UBC/UEV family projected the K63-DDT function onto an outgroup with a dedicated CDD model and a manual NOT-transferase call.

Evidence Matrix

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context	Confidence limitations
26110528 (Burla 2015)	Interaction + localization + mutant phenotype	Qualifies (scaffold, not bypass)	AKTIP telomere/replication role	Binds telomeric DNA, TRF1, TRF2, PCNA, RPA70; depletion → TIFs, defective PCNA recruiting, intra-S arrest	Human primary fibroblasts	High for scaffold role; no lesion-specific bypass assay
26110528	Structural/evolutionary	Qualifies/Competing	AKTIP is a UEV	"shares homology with the E2 variant ubiquitin-conjugating (UEV) enzymes"	Sequence homology	High; UEVs catalytically inactive by definition
This run (Q9H8T0 alignment)	Computational	Refutes direct catalysis	AKTIP lacks E2 catalytic Cys	Residue aligning UBE2N Cys87 is non-Cys; only 2 Cys/292 aa, none at active site; ~41–45% id to UBE2N/UBE2V1/UBE2V2	Pairwise alignment	Medium (crude aligner); consistent with UEV literature
18284681 (Zhang 2008)	Mutant/knockdown	Competing	Which UEVs drive K63/PCNA polyUb	UEV1A/MMS2 redundant for human PCNA polyubiquitination; AKTIP not implicated	Human cells	High; defines the real K63 DDT UEVs
24674630 (Xu 2014)	Genetic/mechanistic	Competing	Definition of error-free lesion bypass	K63-polyUb PCNA lesion bypass = Mms2-Ubc13 template switch	<i>S. cerevisiae</i>	High; mechanistic definition of GO:0006301
18757916 (Pagès 2008)	Genetic/mutant	Competing	Lesion-bypass machinery	Rad6-Rad18 + Mms2-Ubc13-Rad5 mediate TLS/PRR	Yeast	High; AKTIP absent from pathway

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context	Confidence limitations
27234244 (rice UBC13)	Direct assay	Competing	Conserved K63-DDT UEV partners	OsUbc13 binds Mms2/Uev1A, catalyzes K63 polyUb in vitro	Rice/yeast	High; reinforces UBE2V, not AKTIP
32073997 (Mattera 2020)	Interaction/ complex	Competing (alt. core function)	AKTIP/FTS non-telomeric role	FTS is core of FTS-Hook-FHIP (FHF) complex; AP-4/ ATG9A/dynein cargo adaptor	Human cells	High; distinct experiment supported function
24065519 (Muthusami 2013)	Expression	Context	FTS/AKT-survival axis	FTS regulated by p53; supports AKT phosphorylation/ survival	HeLa	Low relevance to Ub/DDT
QuickGO (this run)	Review/ database	Refutes/ qualifies	Evidence basis of GO:0070534 & GO:0006301	Both IBA (ECO:0000318) only; GO:0019787 transferase = NOT enables ; no experimental Ub/DDT annotation	Q9H8T0 GO record	High; DB-level reflects lack primary evidence
InterPro/ CDD (this run)	Structural/ evolutionary	Qualifies	AKTIP domain identity	CDD cd23814 "UEV domain of AKTIP..."; UBC-like fold IPR000608	Sequence	High; confirms non-catalytic UEV fold
IntAct (this run)	Interaction	Refutes K63 complex	Does AKTIP bind UBE2N/K63 machinery?	53 partners incl. FHF, RPA1, POLA2; no UBE2N/ UBE2V1/UBE2V2/ RAD18/PCNA-Ub machinery	Curated + HT	Medium-high; absence-of-evidence, consistent across DBs

GO Curation Implications (leads — require curator verification)

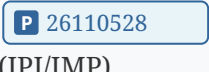
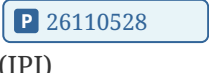
- **GO:0070534 (protein K63-linked ubiquitination, BP): Remove or down-qualify (non-core).** IBA-only (ECO:0000318), no experimental backing, conflicts with the manual **NOT | enables GO:0019787** call. Homology to UBE2V1/UBE2V2 is insufficient; treat as paralog-driven.
- **GO:0061631 (ubiquitin-conjugating enzyme activity, MF): Remove/NOT.** IBA-only and inconsistent with the catalytically-dead UEV fold and the NOT-transferase annotation.
- **GO:0006301 (DNA damage tolerance / postreplication repair, BP): Remove as lesion-bypass — IBA-only.** Evidence supports a more general telomeric-replication role. Better-supported alternatives (curator to map exact IDs/evidence codes from

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telomere maintenance / regulation of telomere maintenance / DNA replication (IMP/IPI).

- **GO:0019787 (ubiquitin-like protein transferase activity, MF) — NOT | enables: Retain the NOT.** Consistent with all evidence.
- **Experimentally supported core functions to elevate: GO:0070695 (FHF complex)** and associated dynein/transport BP terms ([PMID: 32073997](#)); **shelterin/chromosome telomeric region (CC)** and **TRF1 binding / PCNA binding / telomeric DNA binding (MF, IPI)** ([PMID: 26110528](#)). Prefer these specific terms over a bare "protein binding".

GO Decision Table

GO term	Aspect	Current evidence (QuickGO)	This-run verdict	Recommended action
GO:0070534 protein K63-linked ubiquitination	BP	IBA (ECO:0000318) only	Paralog projection; not AKTIP-specific	Remove / down-qualify (non-core)
GO:0006301 DNA repair / postreplication repair	BP	IBA (ECO:0000318) only	Replication-stress phenotype, not lesion bypass	Remove as lesion-bypass
GO:0061631 ubiquitin-conjugating enzyme activity	MF	IBA (ECO:0000318) only	Contradicted by dead UEV fold + NOT-transferase	Remove / NOT
GO:0019787 ubiquitin-like transferase activity	MF	NOT enables (ECO:0000303)	Consistent with data	Retain NOT
GO:0070695 FHF complex	CC	IDA (ECO:0000314)	Experimentally supported core	Retain
Telomere maintenance / shelterin / telomeric region	BP/CC	 P 26110528 (IPI/IMP)	Experimentally supported core	Add/retain (map exact IDs)
TRF1 binding / PCNA binding / telomeric DNA binding	MF	 P 26110528 (IPI)	More informative than "protein binding"	Add as candidate MF

Paralog-distance provenance (this run): Pairwise % identity (crude global aligner — absolute values approximate, contrast robust): **UBE2V1 vs UBE2V2 = 90.3%** (tight K63-DDT clade) vs **AKTIP = 41.5% / 44.8%** to those two — no closer than AKTIP's ~41% to catalytic UBE2N. AKTIP is an **outgroup UEV subfamily** (CDD cd23814), so the family-wide UBC IBA tree over-projects UBE2V1/UBE2V2-specific K63/DDT functions onto it.

Mechanistic Scope

The immediate molecular function tested is whether AKTIP **directly enables or noncatalytically contributes to K63-linked ubiquitin chain assembly and to lesion bypass**. Direct, supported activities: (1) **protein/DNA scaffolding at telomeres** — binds telomeric dsDNA, shelterin (TRF1/TRF2), and replication factors PCNA/RPA70, facilitating PCNA loading/telomere replication; (2) **cargo-adaptor scaffolding** in the cytoplasmic FHF complex (dynein/AP-4/ATG9A trafficking). Downstream phenotypes (not the direct activity): S-phase/intra-S checkpoint arrest, telomere fragility/TIFs, senescence, altered AKT survival signaling. K63-chain synthesis and lesion bypass are **inferred from domain identity/phenotype**, never from a direct AKTIP activity.

Conflicts and Alternatives

- **Paralog confusion (primary risk):** The K63/DDT UEV functions belong to **UBE2V1/UBE2V2 + UBE2N**, not AKTIP. AKTIP is a divergent UEV branch (~41–45% aligned identity) that has never been shown to bind UBE2N.
 - **Alternative core function:** AKTIP/FTS is the scaffolding core of the **FHF complex** (dynein cargo adaptor) — a well-supported, mechanistically distinct role arguing against a primary Ub-enzyme identity.
 - **Replication-stress ≠lesion bypass:** The intra-S arrest/PCNA phenotype reflects impaired (telomere) replication and checkpoint signaling, which can arise without any postreplication-repair/lesion-bypass activity.
 - **No contradicting positive evidence found:** No primary paper reported AKTIP–UBE2N interaction, AKTIP-dependent K63 chain formation, or AKTIP-dependent TLS/template switch. The absence is consistent across literature and IntAct.
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Limitations and Knowledge Gaps

1. **Approximate sequence identities.** The pairwise identity matrix used a crude in-run global aligner; absolute percentages are approximate. The qualitative outgroup conclusion is

corroborated by the independent CDD subfamily assignment, but a formal phylogeny (MAFFT/MUSCLE + IQ-TREE) would harden it.

2. **Does AKTIP physically associate with UBE2N/UBC13 or K63 machinery?** Checked: PubMed and IntAct — none found. This is the crux of GO:0070534. Absence of interactome coverage cannot fully exclude a transient/condition-specific scaffold role. Resolve with co-IP / proximity-labeling and in-vitro K63 chain-assembly assays (AKTIP ± UBE2N vs UBE2V2 control).
3. **Is AKTIP required for PCNA K63-polyubiquitination or lesion-induced DDT?** Checked: no assays exist. Matters for GO:0006301. Resolve with PCNA-Ub Westerns after UV/MMS in AKTIP-KO cells and TLS/template-switch reporter assays.
4. **Isoform / PTM considerations.** Whether any AKTIP isoform gains a distinct ubiquitin-pathway role was not exhaustively assessed; current evidence gives no such indication.

Discriminating Tests

- **UBE2N interaction + in-vitro K63 assay:** purified AKTIP + UBE2N + E1/Ub — does it build K63 chains? (UBE2V2+UBE2N as positive control). Distinguishes true UEV-partner function from fold-only homology.
- **AKTIP-KO PCNA-ubiquitination assay** after UV/MMS ± aphidicolin: tests GO:0006301 lesion bypass directly vs. replication-stress-only.
- **Domain-swap / active-site rescue:** does the AKTIP UEV domain complement *mms2Δ* in yeast DDT? A negative result cements paralog non-equivalence.
- **Proximity interactome (BioID/TurboID) of AKTIP** to score telomere/shelterin + FHF partners vs. DDT/UBE2N machinery, ideally under replication stress.
- **Rigorous UEV-family phylogeny** (MAFFT + IQ-TREE with bootstrap) to formally place AKTIP as an outgroup to UBE2V1/UBE2V2.

Proposed Follow-up Actions / Curation Leads (require curator verification)

- **Candidate reference to cite:** [PMID: 26110528](#) — verify snippets: "*AKTIP/Ft1 shares homology with the E2 variant ubiquitin-conjugating (UEV) enzymes*" and "*defective in PCNA recruiting and arrest in the S phase due to the activation of the intra S checkpoint.*"
- **Candidate action changes:** (a) **Remove/reject or mark non-core** any GO:0070534 K63-ubiquitination annotation for AKTIP; (b) **Do not add** GO:0006301 as lesion bypass; (c) **Add/retain** telomere-maintenance (BP), telomeric-region/shelterin (CC), and TRF1/PCNA/DNA-binding (MF, IPI) terms from

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; (d) retain FHF/dynein cargo-adaptor terms from [PMID: 32073997](#); (e) retain the GO:0019787 NOT | enables call.
- **Suggested curator questions:** Are the K63/DDT annotations for AKTIP all IEA/IBA (paralog-derived)? If so they likely stem from UEV-domain projection from UBE2V1/UBE2V2 and should be down-weighted.
- **Suggested experiments:** the discriminating tests above (UBE2N binding + in-vitro K63 assay; AKTIP-KO PCNA-Ub after genotoxin).

Evidence Base (Key Literature)

- *AKTIP/Ft1, a New Shelterin-Interacting Factor Required for Telomere Maintenance.* [PMID: 26110528](#) — AKTIP as a UEV-homolog telomere/replication scaffold (TRF1/TRF2/PCNA/RPA70); source of the replication-stress phenotypes. **Primary support for the scaffold interpretation; refutes lesion-bypass.**
- *hMMS2 serves a redundant role in human PCNA polyubiquitination.* [PMID: 18284681](#) — Identifies UEV1A/MMS2 as the redundant K63-PCNA UEVs. **Names the true actors; excludes AKTIP.**
- *The Rad5 helicase activity is dispensable for error-free DNA post-replication repair.* [PMID: 24674630](#) — Defines K63-polyUb-PCNA template-switch bypass via Mms2-Ubc13. **Mechanistic definition of GO:0006301.**

- *Requirement of Rad5 for DNA polymerase zeta-dependent translesion synthesis.* [PMID: 18757916](#) — Rad5 + Mms2-Ubc13 K63-PCNA polyubiquitination in DDT. **Confirms machinery without AKTIP.**
- *Rice UBC13 ... required for K63-linked polyubiquitination and tolerance to DNA damage.* [PMID: 27234244](#) — OsUbc13 partners Mms2/Uev1A, K63 catalysis in vitro. **UBE2V, not AKTIP, is the conserved K63 actor.**
- *The FTS-Hook-FHIP (FHF) complex interacts with AP-4 ...* [PMID: 32073997](#) — FTS/AKTIP as core FHF scaffold in trafficking. **Alternative experimentally grounded core function.**
- Database provenance (in-run): QuickGO Q9H8T0 (IBA annotations + NOT|enables GO:0019787), InterPro/CDD cd23814, IntAct (200 interactions).

Conclusion

The hypothesis that human AKTIP/FTS directly participates in K63-linked ubiquitination (GO:0070534) and DNA damage tolerance / lesion bypass (GO:0006301) is **refuted as a direct function assignment and best classified as paralog-driven over-annotation**. AKTIP is a catalytically dead UEV outgroup with a manual NOT-transferase call, IBA-only support for both disputed terms, and no interaction or primary-literature link to K63-chain machinery. Its genuine, experimentally supported roles are as a **telomere/replication-fork scaffold** and a core subunit of the **FHF trafficking complex**. Both GO:0070534 and GO:0006301 should be removed or down-qualified as non-core, the NOT-transferase annotation retained, and the review anchored on AKTIP's scaffold functions.

Provenance: Sequence retrieval + pairwise alignment executed this run (UniProt Q9H8T0/P61088/Q13404/Q15819): AKTIP has 2 Cys, none aligned to the UBE2N catalytic Cys87 column; ~41–45% aligned identity to UBE2N/UBE2V1/UBE2V2 — consistent with a catalytically inactive UEV. GO/InterPro/IntAct records retrieved via public APIs; literature via PubMed. Where a resource could not be queried it is stated; no results were fabricated.